

Course: Engineering Management

Full marks 50

Assessment 50

Total 5 chapters

100% theory

Questions 1 to 6 ab, $15 \times 6 = 90$ and 7abcd

any two $2 \times 5 = 10$ [100 full marks]pass

mark 45

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Unit I: Introduction

(6 Hrs.)

1.1 Management

1.1.1 Functions of management

1.1.2 Level and scope of management

1.1.3 Principles of management

1.2 Organization

1.2.1 Characteristics of organization

1.2.2 Types of organization: formal and informal organizations, virtual organization

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1.3.2 Engineering functions in organizations: product development, operations, IT systems, quality assurance and others

1.3.3 Roles and responsibilities of an engineering manager

Chapter one: Introduction

Meaning of Management

“Management is the art of getting things done through people.”

The major resources of the organization are

- Manpower - Machinery -Materials -Money

Among the entire resources of the organization manpower is very important because all other resources are mobilized and controlled by manpower. The success and the failure of any organization depends up on the manpower. People having different experience, qualification, nature, interest, perception, attitudes are working in different post and position in the organization to achieve the common of the organization.

- So, simply, mobilizing the entire manpower in a systematic way, giving them right task is called management. Researches say that the 85% of the problem of the organization is created by the employees of the organization. So, if the employees are treated well in the organization, then most of the problems are solved. it is an art because the results of management depend upon the personal skill of managers. The art of manager is essential to make the best use of management sciences.

- Management is that field of human behavior in which managers plan, organize, staff, direct, and control human, financial resources in an organized group effort in order to achieve desired individual and group objectives with optimum efficiency and effectiveness.
- As we all know that all organizations have their own goals and objectives. While achieving the goals, organizations perform various activities like planning, staffing, leading and controlling in a systemic and effective way which is known as management.

- Management manages the organizations. It coordinates resources to get jobs done for achieving goals. Resources can be human, finance, materials and information. Management works through and with the people to get jobs done.

- In other words
- Management is a set of activities including planning, leading and controlling directed at organizations resources like human, financial, physical, and information with the aim of achieving organizational goal effectively and efficiently in a changing environment.

Effective: completing assigned task on time

Efficient: less cost, less wastage, quality work and handling machinery properly

- Management gets jobs done through and with the people by using the processes of planning, staffing, leading and controlling to achieve goals efficiently and effectively in a dynamic environment.

- Modern concept of management says that employees must be effective as well as efficient. Effectiveness refers to completion of work or achievement of target on time. Efficiency refers to optimum utilization of resources or no wastage of resources, both are must for every organization; for example, if the annual target of a worker is to produce 1000 units of output, the employee achieves this target by wasting various resources and by mishandling the machinery then the organization will have no benefit so only effectiveness is not required. It must be accompanied by efficiency.

- In the past, management was not considered as an important part of development. With industrial revolution during 17th and 18th century, several economists expressed their 'concepts and function of management'. Only in 19th century, management became the separate field of study because business organization faced various problems regarding labour efficiency and wage payment system. In search of solution of these problems, people began to recognize management as a separate field of study.

Process or Function of management

- Managerial function includes all managerial activities from setting objectives to taking essential steps to ensure and achieve organizational objectives. These functions are essential to create a better working environment to achieve predetermined objectives. The major functions of management include.

Planning:

- planning is the primary function of management. It involves selecting the objectives, policies, procedures and program to achieve a desired result. It is also the process of thinking before doing anything. Since it is a mental exercise and requires intelligence, skill, imagination and vision. Planning gives solution to various problems which may arise in course of functioning. It minimizes future uncertainties and risk. It saves time, effort and cost of the organization.

Planning

- Setting organizational objectives and
- Forecasting the events
- Formulating policies and procedure
- Preparing work schedules and budget

Organizing:

- organizing is the process of identifying the major activities, grouping them into jobs according to the nature and assigning the jobs to different departments and individuals. It is concerned with developing the structure and framework and arranging required resources to perform required activities. These resources include human, financial, physical and information. It defines the authority relationship among the organizational members; it is the root or base of an enterprise. Thus, it is the framework where all mechanisms involved in achieving common objectives are clarified. The major activity of organizing includes
 - Identifying major activities
 - Grouping them into managerial units
 - assigning jobs to different departments and employees
 - delegating necessary authority to fulfill given responsibility

Staffing:

- staffing is concerned with recruitment, selection, appointment and placement of right person to the right job. Staffing is the life blood of an enterprise which mobilizes all other resources for the achievement of common goals. It also involves determining the size of manpower at different levels. Staffing is the continuous process for the continuous existence of an organization. Staffing includes
- Determining the total manpower requirement
- Recruitment, selection and appointment of right person to the right job.
- Organizing seminar, workshop and training to develop employee's skill.
- Performance evaluating promotion and transfer of employees.
- Remunerating employees according to their skill and ability.

Directing:

- Directing is a complex function that includes all those activities which are designed to encourage a subordinate to work effectively and efficiently. It is concerned with instructing, guiding and inspiring subordinates to achieve organizational objectives. Direction is the instrument to develop a better work environment in the organization. The direction function of management includes.

a. Supervision: supervision refers to the direct and immediate guidance to the subordinates to ensure the execution of assigned works. The main motive of supervision is to ensure optimum utilization of human and physical resource so as to achieve organizational objectives. For this, a supervisor has to guide subordinates so that the latter can perform their work electively. It is a continuous process until the existence of an organization.

- **B. Motivation:** motivation is a psychological and human aspect. It is the process of stimulating subordinates to achieve predetermined goals. It is based on need and human behavior. As a social being, workers want to fulfill their basic and social needs. They might be motivated both from financial and non financial incentives.

- **C. Leadership:** leadership is the art of influencing the behavior and performance of the subordinates. It is the ability to persuade others to make them work willingly to achieve the desired goal. A person is said to be a good leader when he is able to influence others and they accept his guidance, suggestions and directions. A manager only if he is able to influence the attitude and behavior of the subordinates.

- **D. Communication:** communication is the process of transmitting ideas and information from one person to another. It is an exchange of facts, opinions, ideas, and views among two or more person. Communication is said to be effective when the message is clear. Effective communication system among all the stakeholders of an organization is essential for its successful operation, so, there must be an effective, direct, and clear communication system to follow information on various management levels.

- **E. Coordination:** coordination is the process of integrating all the units and departments of an organization. It is the process of orderly arrangement of group efforts to provide unity and people perform different functions in an organization. But the activities of different departments are interrelated to each other. Therefore, coordination among all departments is necessary to bring uniformity in action to achieve organizational goal.

Controlling:

controlling is the process of setting a standard, measuring performance, comparing actual performance with that of planned performance and taking corrective action. It is the main instrument with which uniformity in action is maintained. Taking corrective action is necessary if actual performance is not in accordance with the planned performance.

- The management implements a plan through control to ensure the achieving of predetermined goals. Therefore, it is essential to have a good controlling system for achieving results according to the plan. Controlling includes
- Setting standards
- Measuring actual performance
- Identifying deviation, if any, between actual and planned performance
- Analyzing the causes of deviation
- Taking corrective action to achieve predetermined goals.

Needs and importance of management

- Management is essential for the smooth functioning of any organization, whether it is a business, government, non-profit, or even personal life. It ensures that resources are utilized efficiently, goals are achieved, and operations run smoothly. Below are the key **needs** and **importance** of management:

Utilization of resources

Human, finance, materials and information are very important resources of organizations. All these resources are optimally utilized and used by management to achieve the organizational goal in an efficient way. All resources should be equally coordinated to find out the good outputs.

Goal achievement

Achieving goals of organizations is very important thing in management. Management performs various activities like staffing, directing, communicating, planning ect to achieve the goals of the organizations.

Coordination & Teamwork

- Aligns different departments and employees toward a unified purpose.
- Reduces conflicts and improves collaboration.

Adaptability to Change

- Helps organizations respond to market trends, competition, and technological advancements.
- Ensures long-term survival and growth.

Maintaining Order & Discipline

- Establishes rules, policies, and procedures for smooth operations.
- Reduces chaos and improves efficiency.

Increases Efficiency

- Improves productivity by optimizing resource usage.
- Reduces costs and enhances quality and profitability

Facilitates Innovation & Growth

Encourages creativity and new ideas.

- Helps businesses expand and stay competitive.

Ensures Employee Satisfaction

Proper leadership and motivation improve morale.

- Reduces turnover and enhances performance.

Achieves Social & Economic Development

- Businesses contribute to employment, GDP, and societal progress.
- Proper management leads to economic stability and growth.
- Pays tax to the government and helps to increase government revenue.

Customer Satisfaction

- Ensures quality products/services, leading to customer loyalty.
- Builds a strong brand reputation.

Problem solving

There may be various misunderstanding and problems in organizations while performing various activities. Management solves organizational problems. It identifies and analyzes the problems and takes actions to solve it by choosing the best alternative.

- Performance of control

Management sets the standards of work performance in organization. It measures and evaluates the actual performance. It takes corrective action for deviations.

- Management is the backbone of any successful organization. Without effective management, businesses face inefficiency, financial losses, and failure. Good management leads to **higher productivity, employee satisfaction, innovation, and long-term success.**

Scope and application of Management

- Management is a universal concept that applies to all types of organizations—businesses, governments, non-profits, educational institutions, and even personal life. Its **scope** defines the areas where management principles are applied, while its **application** refers to how these principles are implemented in real-world scenarios.
- The **scope of management** refers to the *range of activities, functions, and responsibilities* that managers perform to achieve organizational goals. It basically defines **what management covers** — all the areas and functions that fall under it.

1. Functional Scope (What Managers Do)

Management involves performing several core functions, usually summarized as the **management process**:

1. **Planning** – Deciding what to do, how to do it, and when to do it.
2. **Organizing** – Arranging resources (people, money, materials) to implement the plan.
3. **Staffing** – Recruiting, selecting, training, and developing employees.
4. **Directing** – Guiding, motivating, and leading employees to achieve goals.
5. **Controlling** – Monitoring performance and making corrections if needed.

2. Area Scope (Where Management Applies)

Management applies to different departments or areas in an organization:

1. Financial Management

Budgeting, investment decisions, financial planning, accounting and risk management.

Ensures profitability and sustainability.

2.Human Resource Management (HRM)

Recruitment, training, performance appraisal, and employee welfare.

Ensures a skilled and motivated workforce.

3.Marketing Management

Market research, branding, advertising, sales, and customer relations.

Helps in reaching target customers effectively.

4. Operations Management

Production planning, quality control, supply chain management, and logistics.

Ensures efficient delivery of goods/services.

5. Information Technology (IT) Management

Managing IT infrastructure, cybersecurity, data analytics, and digital transformation.

Supports business efficiency and innovation.

3. Level Scope (Who Manages)

Levels of Management

1. **Top-Level Management** (CEO, Board of Directors)

Strategic planning, policy-making, and long-term goals.

2. **Middle-Level Management** (Department Heads, Managers)

Executing plans, coordinating activities, and supervising lower management.

3. **Lower-Level Management** (Supervisors, Team Leaders)

Direct oversight of employees, day-to-day operations, and task execution.

Application of Management

Management principles are applied in various sectors and scenarios:

A. Business Organizations

- **Corporate Sector:** Managing multinational companies, startups, and SMEs.
- **Retail & E-commerce:** Inventory management, customer service, logistics.
- **Manufacturing:** Production efficiency, quality control, lean management.

B. Public Sector & Government

- **Public Administration:** Policy implementation, governance, and public welfare.
- **Healthcare Management:** Hospital administration, patient care, medical logistics.
- **Educational Management:** School/college administration, curriculum planning.

C. Non-Profit & Social Organizations

- **NGOs:** Fund management, community development, volunteer coordination.
- **Charities:** Resource allocation, donor management, social impact assessment.

D. Personal Life & Self-Management

- Time management, financial planning, career growth, and work-life balance.

E. Emerging Fields

- **Digital & Remote Work Management:** Virtual teams, AI-driven decision-making.

Features of management

- **Management is goal oriented**

Management is a means to achieve predetermined goals. All managerial activities are systematically directed in the accomplishment of such goals. Management has no justification to exist without goals. Effective management is always management by objective. The basic goal of management is to maximize effectiveness and efficient of human efforts. The success of management is measured by the extent to which the established goals are achieved. Hence, management is goal oriented.

- **Management is a group activity**
- Management cannot be done in isolation. It always refers to group efforts. Whenever there is an organized group of people working towards a common goal, some type of management become essential. A group can easily and effectively attain the goal or objectives of an organization rather than individuals.

- **It is a distinct process**
- Management is a distinct process performed to accomplish organizational goals by the use of human and material resource. The chief function of manager is not to do but to get things done through others. Mary Parker Follet has said, management is the art of getting things done through people.

- **Management is a universal activity**
- Management is required in all types of organization. Wherever there is a human activity there is a management. The basic principle of management is applicable in business as well in other organizations. These principles are flexible and are modified according to needs and environment of the organization.

- **Management is an integrating process**
- The essence of management is integration of human and other resources for effective performance. It integrates men, machines, and materials for carrying out the operations for achieving the stated goals. The responsibility of management is welding the three factor men, methods and machinery into a single working factor.

- **Management is a social process**
- Management is a social process because it consists of getting things done through others. This involves dealing with people. The efforts of human being have to be directed, coordinated and regulated by management. Moreover, management has a social obligation to make optimum use of scare resources for the benefit of the community as a whole.

- **Management is both science and art.**
- Management is a science because it has developed certain principles there are universal application, it is also an art because the results of management depend upon the personal skill of managers. The art of manager is essential to make the best use of management sciences. Hence, it is both science and art.

- **Management is a profession**
- Management is now regarded as a profession. It has a systematic and specialized body of knowledge, principles and techniques, which can be taught well as applied in practice. With the increase in joint stock companies, and multinational companies, the management is entrusted in the hands of profession managers.

- **Management is multidisciplinary discipline**
- Management draws knowledge and concepts from other disciplines such as economics, psychology, sociology, statistics; operation research etc. management integrates the ideas and concepts taken from other disciplines and uses them in managing the organization.

- **Management is intangible**
- Management is unseen forces but its results are apparent. The effectiveness of management is judged on the basis of the end result through operations cannot be observed.

- **Management is dynamic**
- Management is a dynamic function and it has to be performed continuously. it operates in an ever-changing environment. Management adapts itself to environmental changes and also introduces innovation in methodology.
- In the past, management was not considered as an important part of development. With industrial revolution during 17th and 18th century, several economists expressed their concepts and function of management. Only in 19th century, management became the separate field of study because business organization faced various problems regarding labor efficiency and wage payment system. In search of solution of these problems, people began to recognize management as a separate field of study.

- **Management is universal**
- Management is applicable to all forms of human organizations whether it is profit- making or non- profit making. Management is not applicable to business undertakings only. It is applicable to political, social, religious and educational institutions also. Management is necessary when group effort is required
- **Management is a social process**
- To get things done from the people manager has to establish inter-personal relations with them. He has to understand the behavior of all as individual members as well as members of a group. Management is a process, function or activity. This process continues till the objectives set by administration are actually achieved.

“Management is a social process involving co-ordination of human and material resources through the functions of planning, organizing, staffing, leading and controlling in order to accomplish stated objectives”.

- **Management is dynamic, and not static**
- Management is not a stereotype activity but is ever changing. It is a complex and dynamic activity. Management adapts itself to changes in environment, and also initiates and introduces changes i.e. Innovations, change in methodology etc.
- Management is a purposeful activity. It coordinates the efforts of workers to achieve the goals of the organization. The success of management is measured by the extent to which the organizational goals are achieved. Management is concerned with directing and controlling of the various activities of the organization to attain the pre-determined objectives.
- Every managerial activity has certain objectives. In fact, management deals particularly with the actual directing of human efforts.

- **Continuous process**
- Management is not one time activity, but it is continuous activity. The cycle of management continues to operate so long as the organizations continue to exist

- **Management is an integrating/unifying force**
- Management is integrating people into a single working force with available physical resources. The essence of management is the integration of human and other resources to achieve the desired objectives. Managers seek to harmonize the individuals' goals with the organizational goals for the smooth working of the organization.

Principles of management

- The practice of management is based on specific disciplines which are known as principles of management. And these management principles are the facts and truths gained through observation and experiment. Simultaneously, various scholars have introduced many principles in different times through their research and investigation. Among them, Henry Fayol has recommended fourteen principles of management.

Division of work:

- this principle is similar to the famous economist, Adam Smith's principle of division of labour. According to this principle, every employee in an organization must be assigned only a specific type of work to increase efficiency. The development of specialization ensures simplicity and accuracy in performance.

Authority and responsibility:

- authority and responsibility are two interrelated terms in management. While authority is the power and right inherent in a managerial position through which a manager commands subordinates, similarly responsibility is the obligation to be fulfilled by the subordinates. This principle emphasizes on the balance between authority and responsibility. Authority without responsibility can make a person irresponsible and there is possibility of misuse of power. Similarly, responsibility without proper authority makes a person ineffective.

Discipline

- Discipline means obedience to superiors and their guidelines. It is also concerned to follow the rules, regulations and procedures of an organization. This principle is applicable to managers and subordinates alike because all are within the rules of the organization. Since the discipline is a must to develop a feeling of self responsibility among the members. The manager has to maintain disciplined environment in the organization.

Unity of command

- : according to this principle every workers/ subordinates must get orders and instructions only from one superior at a time. It means a subordinate should be accountable to a single superior at a time. Further, this principle is essentials to fix the responsibility and avoid confusion. For instance, when a subordinate receives orders and instructions from more than one superior at a time, the will get confused. In such a situation, he cannot fulfill his responsibilities efficiently and may commit mistakes.

Unity of direction

- : unity of direction implies that there should be one head and one plan for a group of activities having a common objective. There must be only plan for common work and one in charge to coordinate all the members of the unit. This unity of direction, thus, is essential to maintain unity in action or operation.

Subordination of individual interest to general interest

- : individual interest means fulfillment of employee's objectives, while general interest means the fulfillment of organizational objectives. This principle concentrates on the basis management philosophy that individual objectives of employees are subordinate to the common objectives because achievement of group objectives in the long run helps to fulfill individual objectives, hence, it is essential to reconcile individual objectives with that of group objectives.

Remuneration of personnel

- : it is the reward paid to the employees for their contribution. The remuneration should be acceptable both to the management and the employees. Wages should be determined by considering employees responsibilities, cost of living and financial condition of the organization. Basic wage should be fixed, beside, there should be a provision of monetary and non monetary incentive.

Centralization and decentralization

- : centralization is the retention of decision making authority at the top level management. On the other hand decentralization is the systematic division of decision making authority from top to bottom in a hierarchy. Maintaining centralization or decentralization depends upon the nature and size of the organizations and also the knowledge and capability of subordinates. Generally, in small organizations where the range of activities is limited, centralization is preferred whereas in large organization decentralization is preferred.

Scalar chain

- : the scalar chain refers to the unbroken line of authority of top to bottom in hierarchy. This principle is important for systematic and orderly communication of information in an organization; information has to be communicated through successive chain from top to bottom or vice versa. According to the Fayol, the scalar chain must be strictly followed except in emergencies. Fayol also has suggested the concept of Gang Plank to avoid delay in communication among the same level of employees.

Order

- : this principle is concerned with the systemic arrangement of materials and people. Fayol has classified order into two types, material order and social order. And placing machines and other physical things in proper place and quantity is material order, and placing right person to the right job is called social order. The material order emphasizes that all physical resources necessary for proper functioning must be available at the right place at the right time. Similarly, social order emphasizes that all necessary members must be available on the proper place. As a result, this principle is essential to minimize the wastage of time and resources.

Equity

- : this principle is concerned with kindness and justice to all the employees working in an organization. Being in the equity, management has to realize that loyalty and devotion can be extracted from the members only through kindness and justice. Obviously, it is helpful in developing a feeling of responsibility among the employees. Therefore, being concerned in it, management has to avoid the concept of favoritism and partiality among the employees.

Stability of tenure

- : according to this principle, employees must have a feeling of security of their job to continue the work efficiently. Instability of employees is the cause of poor management whereas the stability of employees helps to develop experience and efficiency. The feeling of job insecurity among the employees creates problems of high labour turnover, increased cost of selection, training, supervision.

Initiative

- : according to this principle, every employee should be given reliable freedom to exercise judgment in formulation and execution of plans. Obviously, employees do not expect any kind of interruption or guideline from superiors for minor technical work; instead, they want to fulfill their responsibility in their own risk and knowledge. For instance, freedom provides a sense of self motivation among the employees making them more dedicated and loyal towards the organization.

Esprit De Corps

- : Esprit De Corps is proverb that means union is strength. It is possible only through harmony and mutual understanding among the workers. According to this principle, the manager has to take necessary steps to promote team spirit and develop a feeling of harmony among each other. Managers have to attain organizational objectives through group effort. To do so, Fayol suggests two concepts to the managers, firstly, the concept of divide and rule, which much be avoided, and secondly emphasis must be given unity of command and oral communication amount the workers to improve human relations.

- What is Engineering Management?
- Engineering management is the application of the practice of management to the practice of engineering. Engineering management brings together the technological problem-solving ability of engineering and the organizational, administrative, legal and planning abilities of management in order to oversee the operational performance of complex engineering driven enterprises. A Master of Engineering Management (MEM) is sometimes compared to a Master of Business Administration (MBA) .

"The art and science of creating value by using technology together with other resources of an organization is engineering management"

- Engineering management covers the gap between engineering and business management, namely the combination of technical and economic decision-making with analytical skills, optimization capabilities, and technical product development.

- Engineering management skills allow engineers to navigate their work with tact, compassion, and thoughtful strategic planning to boost safety, optimize project finances, make environmentally conscious decisions, support team members, maintain diverse, equitable, and inclusive workplaces, and more.

- In conclusion, engineering management is an art that requires a delicate balance of technical expertise, leadership skills, and strategic thinking.

ICT, or Information and Communication Technology

- ICT, or Information and Communication Technology, refers to the technologies that provide access to information and facilitate communication. This includes hardware (like computers, mobile devices), software (applications and systems), networks (like the internet), and services (such as cloud computing). ICT is a broad field that encompasses many areas, including:
- 1. Telecommunication: Mobile networks, fiber optics, and other forms of communication technology.
- 2. Information Systems: Databases, websites, and enterprise systems.
- 3. Computing: Personal computers, servers, and other computing devices.
- 4. Internet: Email, social media, cloud storage, and web browsing.
- 5. Software Development: Creating programs and applications.
- 6. Networking: Local and wide-area networks, internet access, and network security.
- 7. Artificial Intelligence (AI): Machine learning, data analysis, and automated decision-making.
- ICT plays a crucial role in nearly every aspect of modern life, from business and education to healthcare and government operations.

Concept of organization

- Organization is an association of two or more individuals working together to achieve a common goal; or, in other words, an organization is a collection of people working together in a division of labor to achieve a common objective. When people decide to accomplish any work or job, they need to form an organization. So, it is a social aspect of an essential part of human life. Thus, the concept of organization has been developed from the very beginning of human civilization since people started living in groups of families. Nowadays, people form different types of organizations on the basis of their goals consisting of business, social, political, religious, clubs and so on.

- In fact, a job that cannot be performed by a single person can be performed effectively in a group. So, people used to organizations when they could not perform the job solely in an effective manner. Conclusively, people form an organization on the basis of their common goals, to make division of work on the basis of efficiency, to delegate authority and responsibility, to maintain communication among them, to coordinate the activities among them and to coordinate the activities among all the members. Briefly, an organization promotes collaboration and negotiation among individuals in a group and improves the effectiveness and efficiency in performance to achieve common goals.

- organizations have the structured setting where people work together to achieve common goals. Within the organization there should be clear definition of authority and responsibility among the members. Depending upon the size, organizations complexity have in structure and in division of responsibility and authority. And, they work in a dynamic environment by applying technology to convert inputs into output.

Features of organization

- 1. Collection of people: The concept of organization comes into existence when two or more people come together to accomplish a definite goal. Therefore, organization is a human association in which, people interact with each other to produce a synergic effect and develop a network to communicate information and to maintain unity in work. So, it is essential for effective performance of the enterprise.

- 2. Common goal: The basis of an organization is a specific goal and it serves as a reason of its existence. All the activities of involved members, concentrate on the fulfillment of a common goal. Possibly, it binds the members and motivates them to devote their effort in the best way.

- 3. Division of work: The total work is divided into small units on the basis of their nature in an organization. Each work is assigned to different individuals according to their skills, abilities and experiences. These individualities enable them to develop the concept of specialization and also contribute to develop their working efficiency.

- 4. Coordination: Coordination is the process that integrates the functions of different units of an organization. In an organization, there should be good coordination among the departments and members through executive leadership. Therefore, it is essential to achieve uniform progress of works in each unit. It helps to minimize the duplication and overlap of effort.

- 5. Hierarchy of authority: Organization consists of a formal structure in which hierarchy of authority of each member is clearly defined. The hierarchy of authority is formed on the basis of degree of responsibility and accountability. It clarifies the role of each individual from top to the subordinate levels. Higher level job needs more skill, experience and responsibility by implementing a chain of command to see the work progress of respective subordinates.

- 6. Perpetual existence: An organization is formed for an indefinite period to perform business for a long period of time. An organization continues its existence and operation even while changing in structure, membership, objectives and management. For this, flexibility and adjustment in the line of business in accordance with changing environment is essential. However, a static and unadjustable organization cannot exist for a long period of time.

- 7. environment: an organization performs activities in a dynamic environment. It consumes resources from environment and also exports output to it. The survival and continuance of an organization is assessed in terms of how well it serves in the challenging environment. Basically, impact of external environments like political, economic, socio cultural and technology must be taken into consideration.

- 8. Technology: technology refers to new knowledge, skill, ideas, procedures, equipment and tools. It is essential to convert raw materials into finished products. In an organization people use new technology to perform their activities. The use of appropriate technology in a job helps to develop working efficiency and also helps to minimize the cost of the output.

A virtual organization

- A virtual organization is a business model where geographically dispersed teams and individuals collaborate and function as a unified entity through the use of information and communication technologies (ICT). Unlike traditional organizations, virtual organizations don't have a physical office or centralized location. They rely heavily on digital communication, cloud-based tools, and other online resources to manage operations, projects, and workflows.

- Key characteristics of a virtual organization include:
- 1. Remote Collaboration: Employees or partners work from different locations, often across countries or time zones.
- 2. ICT-Driven: The use of communication tools such as email, video conferencing (e.g., Zoom, Microsoft Teams), instant messaging, and project management platforms (e.g., Slack, Trello) is essential for coordination.

- 3. Flexible Work Environment: There is typically more flexibility in working hours and locations, which can boost work-life balance.
- 4. Cost Efficiency: By eliminating the need for physical office space and other overhead costs, virtual organizations can reduce expenses.

- 5. Global Talent Pool: Virtual organizations can hire talent from anywhere in the world, allowing them to attract highly skilled individuals without being limited by geography.
- 6. Agility and Scalability: virtual organizations can quickly adapt to market changes or scale up/down based on demand.
- Examples of virtual organizations include companies like Gitlab, Automatic, and Buffer, which all operate with fully remote items.

Importance of management in technology driven environments

- Management plays a critical role in technology-driven environments, ensuring that technological innovations are effectively utilized to achieve organizational goals. Here are several reasons why management is vital in such environments:
 - 1. Strategic Alignment
 - Technology Integration with Business Goals: Management ensures that technological investments align with the organization's strategic objectives. This alignment helps drive innovation while keeping the business on course to meet its long-term goals.
 - Foresight in Technology Trends: Good management anticipates future technological trends, helping the organization remain competitive in a rapidly evolving market.

- Strategic alignment is the process of synchronizing an organization's objectives, resources, and operations, achieved through goal alignment, resource allocation, and process coordination. It enables companies to focus their efforts, resources, and talents toward shared purpose and goals.
- Business alignment is a way professionals can arrange the elements of their company to work together to achieve business objectives. Strategic alignment can benefit your company in many ways, including streamlining business systems and improving employee collaboration.

- 2. Resource Allocation
- Efficient Use of Technology: Managers are responsible for allocating financial, human, and technological resources efficiently, ensuring the organization gets maximum value from its tech investments.
- Budget Management: They ensure that technology spending is controlled and aligned with business priorities, avoiding unnecessary or wasteful investments.
- 3. Change Management
- Facilitating Innovation: In a technology-driven environment, change is constant. Management plays a key role in guiding organizations through technological changes, helping employees adapt to new tools and processes.
- Overcoming Resistance: Resistance to change is common, but effective management can minimize this by communicating the benefits and providing training and support.

- 4. Risk Management
 - Cybersecurity: As organizations adopt new technologies, the risk of cyber threats increases. Management is crucial in developing and enforcing security policies, ensuring compliance, and implementing proactive measures to protect the organization.
 - Technology-Related Risks: Managers assess risks related to new tech implementations, such as data loss, system downtime, and technological obsolescence, ensuring proper mitigation strategies are in place.
- 5. Innovation Leadership
 - Fostering Creativity: Management helps create an environment where innovation increases by encouraging experimentation and supporting teams in developing new ideas.
 - Collaborating with Technology Teams: Managers work closely with IT departments, R&D teams, and external tech partners to drive technological innovation that leads to business growth.

- 6. Performance Monitoring
 - Measuring Success: Management monitors the performance of technological solutions to ensure they deliver the expected return on investment (ROI). This involves tracking metrics like productivity improvements, cost savings, and customer satisfaction.
 - Continuous Improvement: By reviewing outcomes, managers can make informed decisions on whether to refine, scale, or discontinue specific technologies or processes.
- 7. Talent Development
 - Training and Upskilling: As technology evolves, employees need continuous training. Management ensures staff have the necessary skills to leverage new tools, driving both individual and organizational success.
 - Attracting Tech: In a technology-driven environment, it's important to attract and retain talent with the technical expertise to drive innovation. Effective management helps create a culture that appeals to such talent.

- 8. Collaboration and Communication
- Cross-Departmental Coordination: Managers facilitate collaboration between technical teams (like IT) and non-technical departments (like marketing or sales) to ensure smooth technology adoption across the organization.
- Global Teams Management: With technology enabling remote work, managers often oversee teams across different geographies, ensuring communication and collaboration tools are used effectively to maintain productivity.

- 9. Customer-Centric Technology Adoption
- Improving Customer Experience: In many cases, technology is adopted to enhance the customer experience. Management plays a key role in ensuring that customer-facing technologies meet customer needs and improve service delivery.
- Data-Driven Decision Making: With access to data analytics tools, management can make better decisions based on customer behavior, market trends, and performance data, ensuring the organization remains responsive and agile.

- In summary, effective management in technology-driven environments is crucial for ensuring that technological innovations are aligned with business strategies, risks are managed, and opportunities for growth and improvement are continuously realized.

Scope of management

- The scope of management refers to the range of activities, tasks, and responsibilities that a manager oversees within an organization. It defines the boundaries within which a manager operates and makes decisions. The scope can vary depending on the level of management (e.g., top, middle, or lower), the size of the organization, and the specific department or function.

- Here are the key components typically included in the scope of management:
- 1. Planning: Setting organizational goals, determining the best strategies to achieve them, and allocating resources.
- 2. Organizing: Structuring the organization, assigning tasks, defining roles, and establishing authority and relationships among employees.

- 3. Leading: Directing and motivating employees to achieve organizational goals. This includes communication, motivation, leadership styles, and conflict management.
- 4. Controlling: Monitoring performance, comparing it with the set standards, and taking corrective actions if necessary.
- 5. Decision-Making: Making choices regarding resources, investments, strategy, staffing, and other organizational needs.

- 6. Human Resource Management: Involves recruiting, training, developing, appraising, compensating, and ensuring the well-being of employees.
- 7. Communication: Ensuring a smooth flow of information both within the organization and externally with stakeholders.
- 8. Resource Allocation: Efficiently allocating resources like time, money, manpower, and technology to maximize productivity and meet organizational objectives.

- 9. Coordination: Ensuring that different departments or teams work together harmoniously to achieve the organization's objectives.
- 10. Risk Management: Identifying, analyzing, and mitigating risks that can affect the organization's objectives.
- Each of these components defines the broad responsibilities of a manager and highlights the scope within which management operates to ensure organizational success.

Why Are Management Skills Important for Engineers?

- As an engineer, you work on complex and expensive projects, use advanced (and often, dangerous) machinery and equipment, collaborate with a diverse team, and influence solutions that have an environmental or social impact. Engineering management skills allow engineers to navigate their work with tact, compassion, and thoughtful strategic planning to boost safety, optimize project finances, make environmentally conscious decisions, support team members, maintain diverse, equitable, and inclusive workplaces, and more.

- As individual contributors, engineers are good at solving technical problems. However, they're often not as strong at communicating the value of technology, leveraging data science techniques to make evidence-based decisions, or leading and inspiring teams of diverse people with empathy and inclusiveness.

- Management skills can help engineers overcome the universal concerns. These skill sets are transferable and applicable in all engineering industries, including mechanical, civil and environmental, aerospace and aeronautical, chemical, biomedical, industrial, and petroleum. An engineer with all-encompassing management skills is able to see the bigger picture. Engineers in leadership roles can identify and prioritize the balance between safety, efficiency, and profit, ensuring a fully rounded project. Engineering managers can connect the technical operation of the technological product or service to its economic outcomes, thereby seeing opportunities that others cannot see.

- Engineering managers are experienced engineers who currently manage technical activities in both large and small organizations. No matter the size of the company, an engineering manager makes decisions that are part of the management chain of command and impact the success of technical activities. Making data-based technical decisions is at the core of what an engineering manager does. The success of an engineering manager's career depends on these decisions. And in today's ever changing technology-based world, these decisions are not easy.

- Thus, engineering management is an in-demand, growing field that's constantly evolving to accommodate modern technical concepts like artificial intelligence (AI) and big data and other technologies emerging from the fourth industrial revolution. Engineering managers are needed in project management, director, and executive roles across engineering fields, including mechanical, civil, environmental, chemical, biomedical, biotech/pharma, industrial, aerospace, aeronautical, and petroleum engineering. Aspiring engineering managers will play a valuable role in the key technical decisions that will help these industries adapt to an ever changing technological world.

- If you aspire to advance your engineering career, an engineering manager role can pave the way for many diverse leadership opportunities. The duties of an engineering manager include planning, designing, and overseeing projects, as well as managing finances and supervising one or more engineering teams. The higher the engineering manager rises up the technical management ladder, the more engineering teams and managers fall under him/her. The work of an engineering manager requires a combination of skills and knowledge including engineering expertise, personnel management, leadership skills, and financial and business acumen.

Engineering functions in organizations

- In an organization, the engineering function plays a crucial role in ensuring the design, development, implementation, and maintenance of systems, products, and processes. Engineering functions vary depending on the industry, but they typically focus on applying scientific and mathematical principles to solve problems and improve efficiency. Here are the primary engineering functions within an organization:

- 1. Research and Development (R&D)
- Purpose: Responsible for innovation, product development, and improving existing products or services.
- Tasks: Conducting research to explore new technologies, creating prototypes, and testing products or systems.
- Example: Developing new materials, product features, or more efficient manufacturing processes.

- 2. Design and Analysis
- Purpose: Creating detailed plans, models, and specifications for new products, systems, or infrastructure.
- Tasks: Designing mechanical systems, electrical systems, software, or infrastructure. Using computer-aided design (CAD) software for simulations and stress testing.
- Example: Designing a new product such as an automobile component or a software system, including analysis of its performance.

- 3. Manufacturing and Production Engineering
- Purpose: Focusing on the efficient production of products or the delivery of services.
- Tasks: Developing and optimizing production processes, improving factory layout, equipment design, and ensuring product quality.
- Example: Setting up assembly lines, ensuring machinery operates at optimal capacity, and reducing waste.

- 4. Quality Control and Assurance
- Purpose: Ensuring that products and processes meet the required quality standards and specifications.
- Tasks: Inspecting materials, parts, and products, testing product durability, and implementing quality management systems.
- Example: Conducting tests to ensure that manufactured products meet international quality standards (ISO).

- 5. Maintenance Engineering
- Purpose: Ensuring that equipment and machinery function effectively with minimal downtime.
- Tasks: Managing preventive maintenance schedules, troubleshooting, and repairing machinery or systems.
- Example: Maintaining machinery in a factory or managing the maintenance of a fleet of vehicles.

- 6. Project Management
- Purpose: Overseeing engineering projects from initiation to completion, ensuring they are delivered on time and within budget.
- Tasks: Coordinating teams, resources, and timelines, as well as managing risks and communication.
- Example: Managing the construction of a new plant or the implementation of new software across a company.

- 7. Technical Support and Service
 - Purpose: Providing support to both internal teams and customers to solve technical problems and improve product performance.
 - Tasks: Troubleshooting equipment, providing training to end users, and implementing upgrades.
 - Example: Offering customer support for complex machinery or assisting with system upgrades in an IT department.
- 8. Process Engineering
 - Purpose: Optimizing processes to increase efficiency, reduce costs, and improve safety.
 - Tasks: Analyzing and improving production processes, implementing lean manufacturing techniques, and ensuring compliance with environmental standards.
 - Example: Reducing energy consumption in manufacturing or redesigning a workflow to minimize bottlenecks.

- 9. Safety and Compliance
- Purpose: Ensuring that engineering projects and operations comply with safety regulations and industry standards.
- Tasks: Developing safety protocols, conducting risk assessments, and implementing corrective actions to prevent accidents.
- Example: Ensuring compliance with occupational health and safety regulations or managing environmental impact assessments.

- 10. Supply Chain and Logistics Engineering
- Purpose: Managing the efficient flow of materials, components, and products from suppliers to the customer.
- Tasks: Designing supply chain systems, improving logistics processes, and optimizing inventory management.
- Example: Implementing just-in-time (JIT) manufacturing practices or improving warehouse automation.

- 11. Systems Integration
- Purpose: Ensuring that different systems, both hardware and software, work together seamlessly.
- Tasks: Coordinating between multiple teams or systems, ensuring that all components of a project are compatible and functional.
- Example: Integrating software platforms in an organization or ensuring different subsystems of an aircraft work together.

- 12. Environmental and Sustainability Engineering
- Purpose: Minimizing the environmental impact of an organization's activities.
- Tasks: Designing eco-friendly processes, reducing emissions, and improving waste management practices.
- Example: Implementing a waste recycling program or reducing water and energy usage in a factory.

- 13. Cost Engineering
- Purpose: Managing and controlling the costs of engineering projects to ensure profitability.
- Tasks: Estimating project costs, analyzing cost-effective alternatives, and optimizing resource allocation.
- Example: Preparing budgets for infrastructure projects and controlling expenses throughout the project lifecycle.

Each of these engineering functions is critical to ensuring an organization's success, whether through innovation, maintaining operational efficiency, or ensuring product quality. Depending on the size and type of organization, engineering roles might overlap or require highly specialized skills.